Automatic Guitar Tuner Specifications Document

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Specifications

1 SCOPE

- **General:** This document describes the requirements and verification processes for the Automatic Guitar Tuner System. This system is a mobile and compact device that can automatically tune the strings of a standard guitar. The system is intended to have adjustable arms that, when mounted on the head of a guitar, can rotate the guitar pegs to tune each individual string. The system uses the sound generated by the strings to determine if the guitar strings are tuned to the correct pitch. The user is expected to generate the sound by plucking each string.

2 ENGINEERING REQUIREMENTS

2.1 Item Description

The Automatic Guitar Tuner System is designed to interact with the mechanical components of a standard guitar in order to produce a tuned guitar (refer to Figure 1).

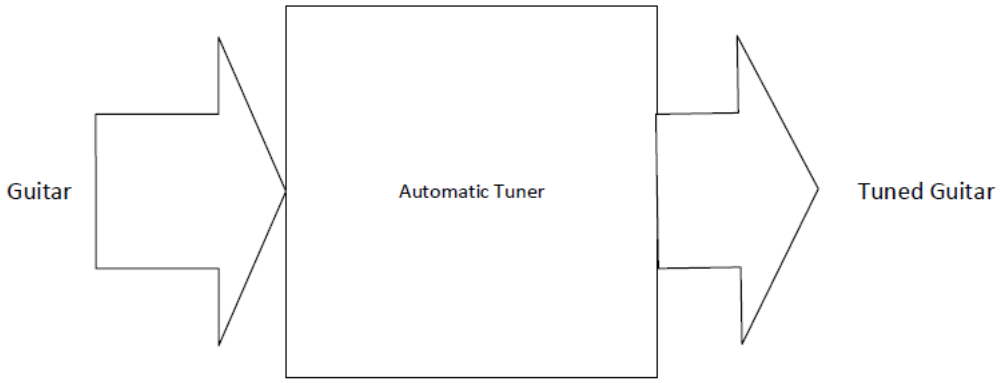


Figure 1: High Level Diagram

To accomplish this function, the system consists of four modularized parts: an input module, a processing module, a controller module, and a mechanical module (refer to Figure 2).

The input of the system will be the sound generated by a given string of the guitar. The output will be the necessary movement required by the mechanical components of the system to adjust the mechanical components of the guitar, namely, the peg corresponding to that string. The

adjusted position of the peg will result in a more accurate pitch produced by the sound of the guitar string.

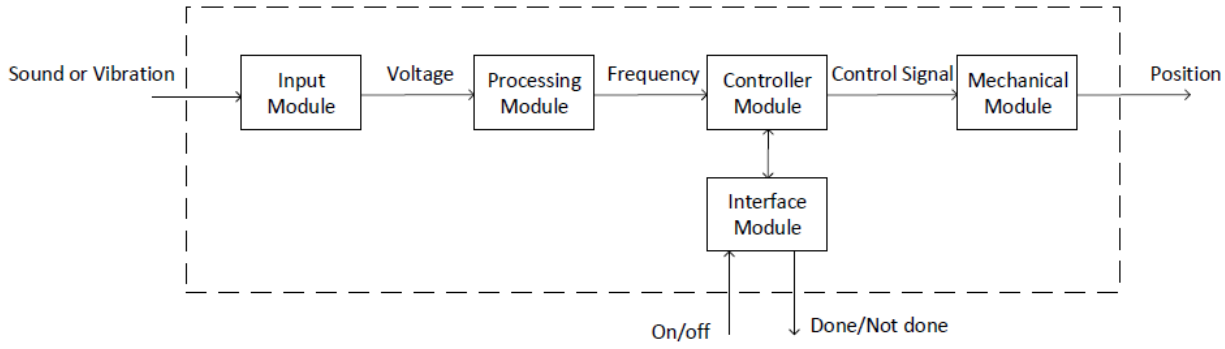


Figure 2: Modularized Functional Diagram

The non-functional parts of the system consist of a power component, physical requirements, and environmental factors (refer to Figure 3).

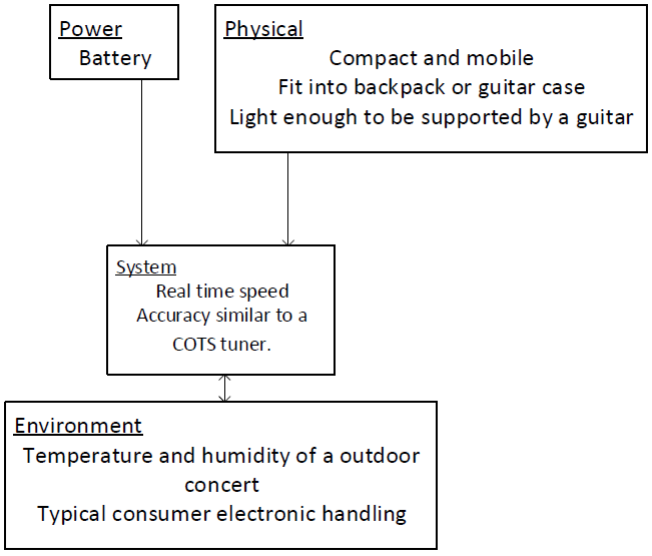


Figure 3: System Functional Support Diagram

The following sections will describe the requirements of the interface, functional, and non-functional components of the Automatic Guitar Tuner System.

2.2 Interface Requirements

2.2.1 Physical

2.2.1.1 The system shall be able to mount on the head of a guitar with thickness varying from 1.5 to 2.5 cm.

2.2.1.2 The mounting components shall not damage the guitar when the system is mounted.

2.2.1.3 The system shall be able to interface with the pegs of typical six-peg guitar shapes: three pegs on each of the two sides of the head or all six pegs on only one side of the head.

2.2.2 Functional

2.2.2.1 The interface module shall be able to be turned on or off by the user.

2.2.2.2 The interface module shall notify the user when the tuning function is complete with a visual or audio signal.

2.3 Functional Requirements

2.3.1 Input Module

2.3.1.1 The input module shall be able to sense the sound output of a guitar string with a frequency range of 82-350 Hz.

2.3.2 Processing Module

2.3.2.1 The processing module shall be able to detect valid input from the input module and discard invalid input. Valid input is defined as sound or vibration that originates from a guitar string. Invalid input is any sound or vibration that originates from a source other than a guitar string.

2.3.2.2 The processing module shall process the valid input and estimate the frequency of the guitar string.

2.3.3 Controller Module

2.3.3.1 The controller module shall use the frequency estimations to calculate desired physical change to mechanical module.

2.3.4 Mechanical Module

2.3.4.1 The mechanical module shall contain at least one mechanism to clasp onto all six pegs of the guitar during operation.

2.3.4.2 When the system is turned on, the mechanical module shall be able to rotate the position of the guitar pegs with an angular range of 0-360°, both clockwise and counterclockwise, in order to adjust the tension of the guitar strings.

2.3.4.3 When the system is turned off, the mechanical module shall cease movement.

2.3.4.4 The torque of the motor(s) used to turn the pegs shall be able to apply a torque of at least 0.3 Nm.

2.3.5 Performance

2.3.5.1 The system shall be able to tune a guitar string to each of the standard six string guitar notes: E2, A3, D3, G3, B4, and E4. The tuning shall be accurate to $+/-$1 cent. A cent is defined as one hundredth of a semitone and can be calculated with the following formula: $c = 1200 * log_2(f_{actual}/f_{desired})$ where $f_{desired}$ is the frequency of the desired note in Hz and f_{actual} is the frequency of the actual note in Hz.

2.3.5.2 The system shall take no more than five seconds to finish tuning a single string.

2.3.6 Error Handling

2.3.6.1 The system shall be able to detect typical errors including: missing strings, broken strings, or improper mounting. When any of these errors are detected, the system shall indicate an error and turn off.

2.3.6.2 The system, when used to tune a guitar, shall not break any guitar strings as long as the string is not over-tuned by one semitones or more. Over-tuned is defined as being tuned to a note that is a higher frequency then the desired note.

2.3.6.3 The system shall not damage the guitar in the case that a string is initially over-tuned and the string breaks.

2.4 Non-Functional Requirements

2.4.1 Electrical

2.4.1.1 The system shall be powered by a battery pack.

2.4.1.2 The battery component shall be able to withstand two weeks of heavy use of the system. Heavy use is defined as one use per day, with each use lasting 30 seconds at most.

2.4.2 Physical

2.4.2.1 All electronic components shall be enclosed in an enclosure which shall fit within a maximum 12 cm x 7.5 cm x 3 cm envelope.

2.4.2.2 The system shall not weigh more than three pounds.

2.4.2.3 The batteries shall be accessible so that they can be replaced.

2.4.3 Environmental

2.4.3.1 The operating temperature range of this system shall be 0 to 40°C.

2.4.4 Economic

2.4.4.1 The system prototype shall have a one-time cost of under $200

2.5 Other Requirements

2.5.1 Safety

2.5.1.1 The system shall not have any exposed conductors.

2.5.1.2 The system shall guard against any pinch points, which are defined as points at which it is possible for a person or part of a person's body to be caught between moving or stationary parts.

3 VERIFICATION OF REQUIREMENTS

3.1 Item Description

Te following sections describe the necessary testing to be performed to verify that the requirements of the Automatic Guitar Tuner System have been met. Each test item corresponds to a requirement of an equivalent enumeration.

3.2 Interface Requirements

3.2.1 Physical

3.2.1.1 Verify that the system is be able to be mounted on the head of a guitar with thickness varying from 1.5 to 2.5 cm. Once mounted, verify that the system remains mounted during use.

3.2.1.2 After mounting the system on a guitar, inspect the guitar and verify that there is no damage on the guitar.

3.2.1.3 Mount the system on two guitars with different head shapes and operate the system as described in the mounting and use instructions. The two head shapes of the guitars are: three pegs on each of the two sides of the head or all six pegs on only one side of the head. Verify that the system is able to rotate the pegs of the two typical six peg guitar shapes

3.2.2 Functional

3.2.2.1 Inspect the system to see if there is a function to turn the system on or off. Turn on the system and observe that the system begins to operate. Turn the system off and verify that it no longer operates.

3.2.2.2 Mount the system on a guitar and operate the system as described in the mounting and use instructions. Wait until the system stops trying to tune the guitar. Verify that the system gives a visual or audio indication that it is done.

3.3 Functional Requirements

3.3.1 Input Module

3.3.1.1 Use the Donner DT-2 tuner and under-tune an E2 guitar string to the frequency of 50 Hz and over-tune an E4 guitar string to the frequency of 350 Hz by manually adjusting

the guitar pegs. Mount the system on a guitar and operate the system as described in the use instructions and verify that the input module can sense these frequencies by observing whether the system can tune the strings.

3.3.2 Processing Module

3.3.2.1 Mount the system on a guitar and turn the system on. Instead of plucking a guitar string to produce an input, generate miscellaneous noises in the vicinity of 5 feet from the system. The miscellaneous noises shall consist of audible voices, clapping, dropping objects of various sizes, and playing music from a speaker.

3.3.2.2 Use the Donner DT-2 tuner and tune a guitar string to the correct pitch by manually adjusting the guitar peg. Then use the tuner to tune another guitar string to the incorrect pitch. Mount the system on a guitar and operate the system as described in the use instructions. Verify that the processing module correctly estimates the frequency output of the guitar by observing that the system adjusts the peg position of the incorrectly tuned string, but the system does not adjust the peg position of the correctly tuned string.

3.3.3 Controller Module

3.3.3.1 Use the Donner DT-2 tuner and over-tune one guitar string one semitone higher and under-tune another guitar string one semitone lower by manually adjusting the guitar pegs of the string. Mount the system on the guitar and operate the system as described in the use instructions. Verify that the system rotates the pegs of the guitar to loosen the tension of the over-tuned string and tighten the tension of the under-tuned string.

3.3.4 Mechanical Module

3.3.4.1 Mount the system on the guitar and operate the system as described in the use instructions. Verify that at least one mechanism is clasping all of the pegs of the guitar.

3.3.4.2 Use the Donner DT-2 tuner and over-tune one guitar string one semitone higher and under-tune another guitar string one semitone lower by manually adjusting the guitar pegs of the string. Mount the system on the guitar and operate the system as described in the use instructions. Verify that the system rotates the guitar peg of the over-tuned string in a full counter-clockwise motion of at least 360°. Verify that the system rotates the guitar peg of the under-tuned string in a full clockwise motion of at least 360°.

3.3.4.3 Mount the system on the guitar and do not turn on the system. Verify that the mechanical components do not move. Operate the system as described in the use

instructions and turn off the system during operation. Verify that the mechanical components cease movements.

3.3.4.4 Examine the specifications of the motors used in the system and verify that they are able to apply more than 0.3 Nm.

3.3.5 Performance

3.3.5.1 Mount the system on a guitar and operate the system as described in the mounting and use instructions. Pluck each string after the system has tried to tune the strings and use a Donner DT-2 tuner to verify if each string is tuned correctly. From the thickest to the thinnest string, the guitar strings should be tuned to the following notes: E2, A3, D3, G3, B4, and E4. Since the Donner tuner has an accuracy of $+/-$ 1 cent, if the tuner indicates that the notes are correctly tuned then the desired accuracy of the system is also verified.

3.3.5.2 Mount the system on a guitar and operate the system as described in the mounting and use instructions. Start a timer when the system starts to tune a string. Stop the timer when the system indicates it is done tuning. Verify that the timer indicates five seconds or less.

3.3.6 Error Handling

3.3.6.1 Take off one of the strings of a guitar. Mount the system on the guitar and operate the system so that it will try to tune the missing string. Verify that the system will stop trying to tune the guitar and indicate that an error has occurred.

3.3.6.2 Use the Donner DT-2 tuner and guitar pegs to tune each string of a guitar one semitone higher then the strings are supposed to (E2 to F2, A3 to B3 flat, etc.). Mount the system on a guitar and operate the system as described in the mounting and use instructions. Verify that none of the strings broke.

3.3.6.3 Use the Donner DT-2 tuner and over-tune a guitar string two semitones higher than the string's intended frequency. Mount the system on a guitar and operate the system as described in the use instructions. Observe that the system tightens the tension of the over-tuned string and breaks the string. Inspect the guitar and verify that there is no damage on the guitar other than the broken string.

3.4 Non-Functional Requirements

3.4.1 Electrical

3.4.1.1 Examine the physical system or available documentation to verify if the systems power source is a battery pack.

3.4.1.2 For two weeks, operate the system for at least 30 seconds every day. Verify that the battery component does not need to be replaced until the end of the two week period.

3.4.2 Physical

3.4.2.1 Examine the physical system and verify that all electronic components are enclosed in an enclosure. Measure the enclosure and verify that fits in an 12 cm x 7.5 cm x 3 cm envelope.

3.4.2.2 Weigh the system and verify that it weighs less than three pounds.

3.4.2.3 Replace the batteries on the system to verify that the battery pack is accessible.

3.4.3 Environmental

3.4.3.1 Mount the system on a guitar and operate the system in an environment with a temperature of 0°C and an environment with a temperature of 40°C. Verify that the system functions according to the defined requirements in both environments.

3.4.4 Economic

3.4.4.1 Calculate the total cost of materials and components required in the construction of the system prototype. Verify that the total cost is less than $200.

3.5 Other Requirements

3.5.1 Safety

3.5.1.1 Inspect the system and verify that the system encloses all conductors so that the user does not come into contact with the conductors while operating the system.

3.5.1.2 Inspect the system and verify that guards are used to eliminate potential pinch points. Mount the system on a guitar and operate the system according to the instructions. Verify that the user or part of the user's body does not get caught between any moving or stationary parts.

Paragraph Number	Test Type	Tester's Name	Pass/Fail	Date